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How to use these notes

These notes follow the official **MINESEC progression sheet** lesson by lesson. Each lesson has its own section, numbered exactly as in the progression, so you always know where you are in the syllabus.

Feature	What to do with it
Learning objectives (green box)	Read them first: they tell you what you must be able to do by the end of the lesson.
Lesson notes and Key Points (blue box)	Learn the definitions and formulas in the Key Points by heart.
Exam tips (yellow box)	Common mistakes and how GCE examiners expect answers to be set out.
Worked examples	Cover the solution, try the question yourself, then compare line by line.
Practice questions	Do them without looking at the answers. Mark your work and redo any you got wrong.
Review exercise	Mixed exam-style questions to test yourself on several lessons at once.

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Chapter 1: Exploring AI Concepts and Emerging Technologies

Lower Sixth introduced artificial intelligence. This chapter goes further: the ethics of AI and how to use it responsibly, the main AI techniques and intelligent systems, machine learning in more depth, and the stages of developing a real AI system.

Lesson	Title (MINESEC progression)
1	Introduction to Artificial Intelligence (AI) - AI Ethics and Responsible Use
2	AI Techniques and Intelligent Systems
3	Machine Learning
4	Developing AI Systems
5	Integration Activity

Lesson 1: Introduction to Artificial Intelligence (AI) - AI Ethics and Responsible Use

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ explain the principles of responsible AI
- ✔ identify sources of bias and other ethical issues in AI systems

KEY POINTS

Artificial intelligence (AI) is the ability of a computer system to perform tasks that normally need human intelligence: learning, reasoning, recognising speech and images, understanding language and making decisions.

AI now helps to decide who gets a loan or a job interview, which posts we see, how patients are diagnosed and how cars drive. Because these decisions affect people's lives, AI must be built and used **responsibly**.

Principles of responsible AI

Principle	Meaning	Example of a failure
Fairness	The system must not discriminate against any group	A hiring tool trained on past (mostly male) employees rejects women
Transparency and explainability	People should know when AI is used and be able to understand why a decision was made	A loan is refused and no one can say why
Privacy and data protection	Collect only the data needed, with consent, and keep it secure	A face-recognition app stores photos without permission
Accountability	A human or organisation remains responsible for the AI's decisions	"The computer decided" is used to avoid responsibility
Safety and security	The system must be reliable and protected against misuse and attacks	A chatbot gives dangerous medical advice
Human oversight	Important decisions are reviewed by people who can override the AI	A self-driving system with no way for a human to take control

Sources of bias

KEY POINTS

Algorithmic bias happens when an AI system gives unfair results for some groups. It usually comes from **biased training data** (data that under-represents a group or reflects past discrimination), from poorly chosen features, or from how results are used.

Ethical issues and misuse

- **Deepfakes:** realistic fake videos, images or voices used for fraud, blackmail or false news.
- **Misinformation:** AI can generate convincing but false text very quickly.
- **Surveillance:** face recognition can track people without their knowledge.
- **Jobs:** automation can replace some jobs, while creating new ones (data analysts, AI trainers).
- **Intellectual property:** AI trained on other people's work raises questions of copyright.
- **Over-reliance:** users may trust AI answers without checking them; AI can "hallucinate" (state false facts confidently).

Responsible use by students

- Use AI tools to **learn and check**, not to copy answers; always verify facts in reliable sources.
- Never upload private data (yours or others') to AI tools.
- Say when you have used AI in your work if your teacher or exam rules require it.

Example: A bank uses AI to approve loans. Applicants from one region are refused far more often than others with similar incomes. Identify the ethical problem and suggest two actions.

> The problem is **unfairness (bias)**: the system discriminates by region, probably because of biased training data.
>
> Actions: test the model's results for each group and retrain it on balanced data; make decisions explainable and let a human officer review refusals.

Practice Questions: Lesson 1

1. State two principles of responsible AI and explain each.
2. A face-recognition system works well on light-skinned faces but poorly on dark-skinned faces. Name the problem and give its probable cause.

Answers: *1. E.g. fairness (no discrimination against any group); privacy (collect only necessary data with consent and protect it) 2. Algorithmic bias, caused by training data that under-represents dark-skinned faces*

Lesson 2: AI Techniques and Intelligent Systems

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ describe the parts of an expert system and its reasoning methods
- ✔ describe common AI techniques and intelligent agents

Knowledge-based (expert) systems

KEY POINTS

An **expert system** copies the decision-making of a human expert in a narrow field. Its parts: a **knowledge base** (facts and IF–THEN rules), an **inference engine** (applies the rules to the facts), a **user interface** and an **explanation facility** (shows how it reached a conclusion).

Forward chaining	Backward chaining
Starts from the known facts	Starts from a goal (hypothesis)
Applies rules to reach new conclusions	Looks for rules and facts that prove the goal
"Data-driven"	"Goal-driven"
Used for monitoring and planning	Used for diagnosis

Example: Rules: R1: IF fever AND rash THEN measles. R2: IF measles THEN isolate patient. Facts: fever, rash. Use forward chaining.

From fever and rash, R1 gives "measles"; from measles, R2 gives "isolate patient".

Search techniques

- **Breadth-first search (BFS):** explores all neighbours of a node before going deeper; finds the shortest path in steps.
- **Depth-first search (DFS):** goes as deep as possible along one path before backtracking; uses little memory.
- **Heuristic search (e.g. A*):** uses an estimate of the distance to the goal to choose the most promising path first; used in route planning (maps) and games.

Other techniques

Technique	Idea	Example
Fuzzy logic	Degrees of truth between 0 and 1 instead of only true/false	Air conditioner adjusting for "slightly warm"
Neural networks	Layers of connected artificial neurons whose weights are learned from data	Handwriting recognition
Genetic algorithms	Solutions "evolve" by selection, crossover and mutation	Timetable optimisation
Natural language processing (NLP)	Understanding and generating human language	Chatbots, translation
Computer vision	Understanding images and video	Face unlock, number-plate readers
Generative AI	Large models create new text, images or sound	ChatGPT, image generators

Intelligent agents

> **KEY POINTS**
> An **intelligent agent** perceives its environment through **sensors** and acts on it through **actuators** to achieve its goals.

Example: Identify the sensors, actuators and goal of a robot vacuum cleaner.

Sensors: bump sensors, cliff (drop) sensors, camera or lidar, dirt sensor. Actuators: wheels, brushes, suction motor. Goal: clean the whole floor efficiently without falling or getting stuck.

Practice Questions: Lesson 2

1. Name the four parts of an expert system.
2. Identify one sensor and one actuator of a self-driving car.

Answers: *1. Knowledge base, inference engine, user interface, explanation facility 2. Sensor: camera, radar or lidar; actuator: steering motor, brakes or accelerator*

Lesson 3: Machine Learning

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ distinguish the three types of machine learning
- ✔ explain overfitting, test sets and accuracy

KEY POINTS

Machine learning (ML) is a branch of AI in which a computer learns patterns from **data** instead of following rules written by a programmer. The result of training is a **model** that can make predictions on new data.

Traditional programming	Machine learning
Input: data + rules	Input: data + correct answers (examples)
Output: answers	Output: the rules (a model)

Types of machine learning

Type	Data	Typical task	Example
Supervised	Labelled (inputs with correct outputs)	Classification (a category), regression (a number)	Spam detection; predicting house prices
Unsupervised	Unlabelled	Clustering, association	Grouping customers; "people who bought X also bought Y"
Reinforcement	Rewards and penalties from an environment	Sequences of decisions	Game-playing agents; robot walking

Key terms

KEY POINTS

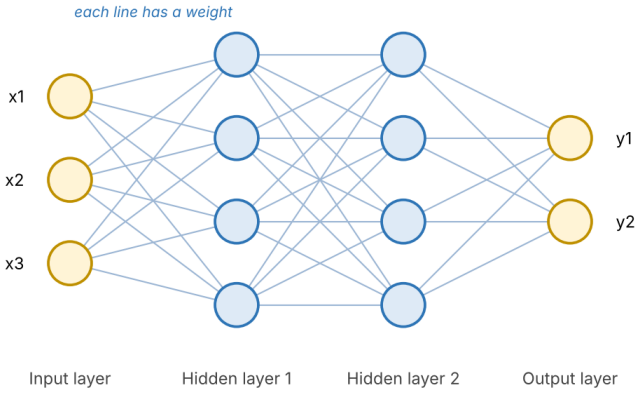
Features: the input variables (e.g. size, location). **Label:** the output to predict (e.g. price).

Training set: data used to learn. **Test set:** separate, unseen data used to measure how well the model works.

Overfitting: the model memorises the training data (very good on training data, poor on new data). **Underfitting:** the model is too simple to learn the pattern (poor on both).

Accuracy = number of correct predictions ÷ total number of predictions.

Neural networks and deep learning



A neural network has an **input layer**, one or more **hidden layers** and an **output layer**. Each connection has a **weight**; each neuron adds its weighted inputs and applies an **activation function**. Training adjusts the weights to reduce the error (back-propagation). **Deep learning** uses many hidden layers.

Example: A spam filter is tested on 250 emails and classifies 230 correctly. Find its accuracy. It scored 99% on its training data. What does this suggest?

Accuracy = 230 ÷ 250 = 0.92 = 92%.

99% on training data but 92% on test data suggests slight **overfitting**.

Example: A simple neuron has inputs x1 = 1, x2 = 0, x3 = 1, weights 0.4, 0.9, −0.2 and outputs 1 if the weighted sum ≥ 0.1, otherwise 0. Find the output.

Sum = 1(0.4) + 0(0.9) + 1(−0.2) = 0.2. Since 0.2 ≥ 0.1, the output is 1.

Practice Questions: Lesson 3

1. Which type of machine learning groups customers without labels?
2. A model is correct 171 times out of 180 test cases. Find its accuracy.

Answers: *1. Unsupervised learning (clustering) 2. 171 ÷ 180 = 0.95 = 95%*

Lesson 4: Developing AI Systems

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ outline the stages of developing an AI system
- ✔ explain the importance of good data

Stages of developing an AI system

1. **Define the problem:** what must be predicted or decided, and how success will be measured.
2. **Collect data:** from databases, sensors, surveys, open datasets.
3. **Prepare the data:** clean it (remove errors, duplicates and missing values), label it, and split it into training and test sets.
4. **Choose a model:** decision tree, neural network, k-nearest neighbours, etc.
5. **Train** the model on the training data.
6. **Evaluate** it on the test data (accuracy and other measures); improve it if necessary.
7. **Deploy** it in an app, website or device.
8. **Monitor and retrain** it as new data arrives.

The importance of data

KEY POINTS

"**Garbage in, garbage out**": a model is only as good as its data. Data must be **sufficient** (enough examples), **accurate**, **relevant**, **representative** (covers all the groups the system will be used on) and **legally and ethically obtained**.

- **Data cleaning:** fixing errors and missing values.
- **Data labelling:** adding the correct answers (often done by people).
- **Data split:** a common split is 80% training and 20% testing.

Tools

- **Python** with libraries: NumPy and pandas (data), scikit-learn (classic ML), TensorFlow and PyTorch (deep learning).
- **Jupyter notebooks**, Google Colab; no-code tools such as Teachable Machine.

```
# Training a simple classifier with scikit-learn
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier

X, y = load_iris(return_X_y=True)                # features and labels
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
model = DecisionTreeClassifier()
model.fit(X_train, y_train)                      # training
print("Accuracy:", model.score(X_test, y_test))  # evaluation
```

Example: A dataset has 5 000 labelled images. How many are used for training and testing with an 80/20 split?

Training: 0.8 × 5000 = 4000 images; testing: 1000 images.

Practice Questions: Lesson 4

1. State the stage that follows "train the model".
2. What does "garbage in, garbage out" mean for AI?

Answers: *1. Evaluate the model on test data 2. A model trained on poor, wrong or biased data gives poor, wrong or biased results*

Lesson 5: Integration Activity

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ apply AI concepts to a real situation

Lesson notes

Situation: A hospital in your town wants an AI system that reads chest X-rays and warns doctors about possible pneumonia. The director asks your class for advice.

Worked examples

Example: Task 1. Which type of machine learning is suitable and why?

Supervised learning (classification): the system learns from X-rays labelled "pneumonia" or "normal" by doctors.

Example: Task 2. List the stages the team should follow to develop the system.

Define the problem and success measure → collect X-rays → clean, label and split the data → choose a model (e.g. a deep neural network) → train → evaluate on test images → deploy in the hospital → monitor and retrain.

Example: Task 3. The training images all come from adults in one city hospital. What risk does this create?

The data is **not representative**: the system may be less accurate for children or patients from other areas (**bias**). More varied data should be collected.

Example: Task 4. On 400 test X-rays the system is correct 372 times. Find the accuracy.

372 ÷ 400 = 0.93 = 93%.

Example: Task 5. State three responsible-AI rules the hospital must follow.

Patient data is kept private and used with consent; a doctor reviews every AI warning (human oversight); the hospital remains accountable and can explain how the tool is used.

Practice Questions: Lesson 5

1. Why must the X-ray dataset include children as well as adults?
2. Give one way the hospital keeps a human in control of the AI.

Answers: *1. So that the model is representative and accurate for children too (avoids bias) 2. A doctor reviews every AI warning before any decision*

Review Exercise: Exploring AI Concepts and Emerging Technologies

Answer these mixed questions after finishing the topic. They combine ideas from several lessons, like GCE questions do.

Review Questions

1. State and explain four principles of responsible AI.
2. Explain how biased training data can lead to unfair AI decisions. Give an example.
3. Distinguish between forward chaining and backward chaining.
4. Name the three types of machine learning and give one example of each.
5. Explain overfitting and how a test set helps to detect it.
6. A model makes 450 correct predictions out of 500. Calculate its accuracy.
7. Describe the stages of developing an AI system.
8. Give four qualities of a good dataset for training an AI model.

Answers: *1. E.g. fairness (no discrimination), transparency (decisions can be explained), privacy (data collected with consent and protected), accountability (people remain responsible) 2. A model learns the patterns in its data; if a group is missing or past decisions were unfair, the model repeats that unfairness, e.g. a hiring tool trained mostly on men rejecting women 3. Forward chaining starts from known facts and applies rules to reach conclusions (data-driven); backward chaining starts from a goal and looks for facts and rules that prove it (goal-driven) 4. Supervised: spam detection; unsupervised: grouping customers; reinforcement: a game-playing agent 5. Overfitting: the model memorises training data and does badly on new data; a separate test set of unseen data shows the drop in accuracy 6. 450 ÷ 500 = 90% 7. Define the problem, collect data, prepare (clean, label, split) it, choose a model, train, evaluate, deploy, monitor and retrain 8. Sufficient, accurate, relevant, representative (and legally obtained)*

END OF FREE SAMPLE

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